F70LA Life Insurance Mathematics A (2025/26)
Tutorial 8 Solutions

1. The sum assured at each time is as shown. Policy year n corresponds to the time interval $[n-1, n)$, for $n = 1, 2, \ldots, 15$.

Time n	Death benefit (£) in time interval $[n, n+1)$	Survival benefit (£) at time n
0	100,000	
1	106,000	
2	112,000	
⋮	⋮	
14	184,000	
15	N/A	190,000

Note that the sum assured increases at the end of each policy year by £6,000. If the policyholder survives the 15 years, he is paid £190,000. Whereas, if he died the day before that time, he would get £184,000.

(a) The death benefit if the man dies in the first policy year is £100,000.

(b) The death benefit if the man dies in the second policy year is £106,000.

(c) The death benefit if the man dies in the last policy year is £184,000.

(d) The survival benefit at age 60 is £190,000.

(e) Finally, calculate the expected present value of the endowment insurance using the basis AM92 Ultimate at 4% per annum effective (see the last pages of the tutorial sheet for the life table).

The death benefit in year k (with year k being the time interval $[k-1, k)$) is

$$100,000 + (k-1) \times 6,000 = 94,000 + 6,000k.$$

Thus the death benefit can be valued as $94,000\, A^{1}_{45:\overline{15}|} + 6,000\, (IA)^{1}_{45:\overline{15}|}$.
As the present value of the survival benefit is $190,000\, {}_{15}p_{45}\, v^{15}$, the total EPV of the insurance benefits is

$$\begin{aligned} &94,000\, A^{1}_{45:\overline{15}|} + 6,000\, (IA)^{1}_{45:\overline{15}|} + (94,000 + 96,000)\, {}_{15}p_{45}\, v^{15} \\ =&94,000\, A_{45:\overline{15}|} + 6,000\, (IA)^{1}_{45:\overline{15}|} + 96,000\, {}_{15}p_{45}\, v^{15}. \end{aligned}$$

Using AM92 Ultimate at 4% per annum effective,

$$A_{45:\overline{15}|} = 0.56206,$$

$$ {}_{15}p_{45}\, v^{15} = \frac{\ell_{60}}{\ell_{45}}\,(1.04)^{-15} = \frac{9287.2164}{9801.3123}\,(1.04)^{-15} = 0.52614 $$

and

$$ \begin{aligned} (IA)^{1}_{45:\overline{15}|} =& (IA)_{45} - {}_{15}p_{45}\, v^{15}\ ((IA)_{60} + 15A_{60}) \\ =& 8.33628 - (0.52614)\ (8.36234 + 15 \times 0.45640) \\ =& 0.33457. \end{aligned} $$

Thus the total EPV of the insurance benefits is

$$ 94,000\,(0.56206) + 6,000\,(0.33457) + 96,000\,(0.52614) = £105,350.50. $$

2. The sum assured at each time is as shown. Policy year n corresponds to the time interval $[n-1, n)$, for $n = 1, 2, \ldots, 20$.

Time n	Death benefit (£) in time interval $[n, n+1)$	Survival benefit (£) at time n
0	$100,000$	
1	$100,000 \times 1.0923$	
2	$100,000 \times 1.0923^2$	
$\vdots$	$\vdots$	
19	$100,000 \times 1.0923^{19}$	
20	N/A	$100,000 \times 1.0923^{20}$

   (a) The death benefit if the man dies during the first policy year is £100,000.

   (b) An expression for the death benefit if the man dies during the second policy year is $£100,000 \times 1.0923^1$.

   (c) An expression for the death benefit if the man dies during the last policy year is $£100,000 \times 1.0923^{19}$.

   (d) An expression for the survival benefit if the man survives to age 65 is $£100,000 \times 1.0923^{20}$.

   (e) Finally, we calculate the expected present value of the endowment insurance using the basis AM92 Ultimate at 6% per annum effective.

   This is a geometrically increasing benefit. To value it, go back to first principles.

   The EPV of the death benefit (20-year term insurance) is

$$ \begin{aligned} & 100,000\left(v\,q_{45} + v^2\,(1.0923)\,{}_{1|}q_{45} + v^3\,(1.0923)^2\,{}_{2|}q_{45} + \cdots + v^{20}\,(1.0923)^{19}\,{}_{19|}q_{45}\right) \\ =& \frac{100,000}{1.01923}\ \left((1.0923\,v)\,q_{45} + (1.0923\,v)^2\,(1.0923)\,{}_{1|}q_{45} + \cdots + (1.0923\,v)^{20}\,{}_{19|}q_{45}\right). \end{aligned} $$

Note that $(1.0923\,v) = \frac{1.01923}{1.06} = \frac{1}{1.04}$, so the EPV is

$$\begin{aligned}&\frac{100,000}{1.01923}\left((1.04)^{-1}\,q_{45} + (1.04)^{-2}\,{}_{1|}q_{45} + (1.04)^{-3}\,{}_{2|}q_{45} + \cdots + (1.04)^{-20}\,{}_{19|}q_{45}\right)\\ =&\frac{100,000}{1.01923}\,A^{1}_{45:\overline{20|}}, \qquad \text{valued at } i = 4\% \text{ p.a.}\end{aligned}$$

To determine the value from the Tables, first note $A^{1}_{45:\overline{20|}} = A_{45:\overline{20|}} - v^{20} \times {}_{20}p_{45}$. As

$$v^{20} \times {}_{20}p_{45} = (1.04)^{-20} \times \frac{\ell_{65}}{\ell_{45}} = (1.04)^{-20} \times \frac{8821.2612}{9801.3123} = 0.41075198$$

and, reading off from the AM92 4% tables, $A_{45:\overline{20|}} = 0.46998$, we get

$$A^{1}_{45:\overline{20|}} = A_{45:\overline{20|}} - v^{20} \times {}_{20}p_{45} = 0.46998 - 0.41075 = 0.05923.$$

Hence the EPV of the death benefit (20-year term insurance) is

$$\frac{100,000}{1.01923}\,A^{1}_{45:\overline{20|}} = \frac{100,000}{1.01923} \times 0.05923 = 5,811.250$$

The EPV of the survival benefit (20-year pure endowment) is (using the calculation of $v^{20} \times {}_{20}p_{45}$ at 4% p.a. above)

$$\begin{aligned}100,000 \times 1.0923^{20} \times (1.06)^{-20} \times {}_{20}p_{45} =& 100,000 \times (1.04)^{-20} \times {}_{20}p_{45}\\ =& 100,000 \times 0.41075198\\ =& 41,075.198.\end{aligned}$$

The EPV of the 20-year endowment insurance is

$$5,811.250 + 41,075.198 = £46,886.45.$$

3. (a)

$$\begin{aligned}\ddot{a}_{80:\overline{5|}} =& 1 + \frac{v\ell_{81} + v^2\ell_{82} + v^3\ell_{83} + v^4\ell_{84}}{\ell_{80}}\\ =& 1 + \frac{(1.04)^{-1}(52\,293) + (1.04)^{-2}(48\,858) + (1.04)^{-3}(45\,332) + (1.04)^{-4}(41\,736)}{55\,607}\\ =& 4.0829.\end{aligned}$$

(b) Using Woolhouse's formula,

$$\ddot{a}^{(12)}_{80:\overline{5|}} \approx \ddot{a}_{80:\overline{5|}} - \frac{11}{24}\left(1 - A_{80:\overline{5|}}\right) = 4.0829 - \frac{11}{24}\left(1 - 0.562875\right) = 3.8826.$$

4. The basis is the AM92 survival model and 4% per year interest.

(a) Using Woolhouse's formula,

$$\ddot{a}_{70}^{(4)} \approx \ddot{a}_{70} - \frac{3}{8} = 10.375 - 0.375 = 10.000$$

(b) Similarly,

$$\ddot{a}_{80}^{(4)} \approx \ddot{a}_{80} - \frac{3}{8} = 6.818 - 0.375 = 6.443.$$

(c) As

$${}_{10}p_{70}v^{10} = \frac{\ell_{80}}{\ell_{70}} \times (1.04)^{-10} = \frac{5266.4604}{8054.0544} \times (1.04)^{-10} = 0.44174$$

and using the Woolhouse formula above,

$$\ddot{a}_{70:\overline{10}|}^{(4)} = \ddot{a}_{70}^{(4)} - {}_{10}p_{70}\, v^{10}\, \ddot{a}_{80}^{(4)} \approx 10.000 - (0.44174)(6.443) = 7.154.$$

(d) For the continuously-paid term life annuity, first calculate

$$\bar{a}_{70} \approx \ddot{a}_{70} - \frac{1}{2} = 9.875 \quad \text{and} \quad \bar{a}_{80} \approx \ddot{a}_{80} - \frac{1}{2} = 6.318.$$

Then

$$\bar{a}_{70:\overline{10}|} = \bar{a}_{70} - {}_{10}p_{70}\, v^{10}\, \bar{a}_{80} \approx 9.875 - (0.44174)(6.318) = 7.084.$$

(e) Finally,

$$\bar{A}_{70:\overline{10}|} = 1 - \delta\, \bar{a}_{70:\overline{10}|} \approx 1 - \ln(1.04)(7.084) = 0.7222.$$

5. (a)

$$\ddot{a}_{60} \approx \bar{a}_{60} + \frac{1}{2} = 11.128, \quad \ddot{a}_{90} \approx \bar{a}_{90} + \frac{1}{2} = 3.189, \quad v^{30}\, {}_{30}p_{60} = (1.04)^{-30}(0.03861) = 0.01190.$$

Then

$$\ddot{a}_{60:\overline{30}|} = \ddot{a}_{60} - v^{30}\, {}_{30}p_{60}\, \ddot{a}_{90} \approx 11.128 - (0.01190)(3.189) = 11.090.$$

(b)

$$A_{60:\overline{30}|} = 1 - d\, \ddot{a}_{60:\overline{30}|} \approx 1 - (1 - 1.04^{-1})(11.090) = 0.5735.$$

6. (a) Using Woolhouse's formula and the fact that $\ddot{a}_{x:\overline{1}|} = 1$ and $A_{x:\overline{1}|}^{\,1}$ is a payment upon survival to age $x+1$,

$$\ddot{a}_{x:\overline{1}|}^{(m)} \approx \ddot{a}_{x:\overline{1}|} - \frac{m-1}{2m}\left(1 - A_{x:\overline{1}|}^{\,1}\right) = 1 - \frac{m-1}{2m}\left(1 - vp_x\right) = \frac{m+1}{2m} + \frac{m-1}{2m}\, v\, p_x$$

(b) (i) Consider an annuity payable m-thly in advance with compound annual increases at rate r, so that the rate of payment throughout the nth year is $(1+r)^{n-1}$ per year. The EPV of this annuity is

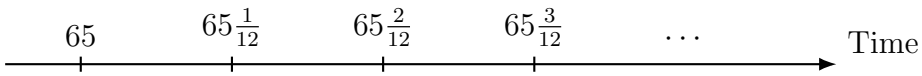
$$\begin{aligned}
&\ddot{a}_{x:\overline{1}|}^{(m)} + (1+r)\, {}_1|\ddot{a}_{x:\overline{1}|}^{(m)} + (1+r)^2\, {}_2|\ddot{a}_{x:\overline{1}|}^{(m)} + \cdots \\
=&\ddot{a}_{x:\overline{1}|}^{(m)} + (1+r)\, v\, p_x \ddot{a}_{x+1:\overline{1}|}^{(m)} + (1+r)^2\, v^2\, {}_2p_x \ddot{a}_{x+2:\overline{1}|}^{(m)} + \cdots \\
=&\sum_{n=0}^{\infty} (1+r)^n\, v^n\, {}_np_x\, \ddot{a}_{x+n:\overline{1}|}^{(m)}.
\end{aligned}$$

(ii) Applying the approximation in part (a) to the terms $\ddot{a}^{(m)}_{x+n:\overline{1}|}$ in the EPV,

$$\begin{aligned}
&\sum_{n=0}^{\infty}(1+r)^n\, v^n\, {}_np_x\, \ddot{a}^{(m)}_{x+n:\overline{1}|} \\
=&\ddot{a}^{(m)}_{x:\overline{1}|} + (1+r)\, v\, p_x \ddot{a}^{(m)}_{x+1:\overline{1}|} + (1+r)^2\, v^2\, {}_2p_x \ddot{a}^{(m)}_{x+2:\overline{1}|} + \cdots \\
\approx&\left(\frac{m+1}{2m} + \frac{m-1}{2m}\, v\, p_x\right) + (1+r)\, v\, p_x \left(\frac{m+1}{2m} + \frac{m-1}{2m}\, v\, p_{x+1}\right) \\
&\qquad + (1+r)^2\, v^2\, {}_2p_x \left(\frac{m+1}{2m} + \frac{m-1}{2m}\, v\, p_{x+2}\right) + \cdots \\
\approx&\frac{m+1}{2m}\left(1 + (1+r)\, v\, p_x + (1+r)^2\, v^2\, {}_2p_x + \cdots\right) \\
&\qquad + \frac{m-1}{2m}\left(v\, p_x + (1+r)\, v^2\, {}_2p_x + (1+r)^2\, v^3\, {}_3p_x + \cdots\right) \\
\approx&\frac{m+1}{2m}\ddot{a}^{\star}_x + \frac{m-1}{2m}\frac{1}{1+r}\left((1+r)\, v\, p_x + (1+r)^2\, v^2\, {}_2p_x + (1+r)^3\, v^3\, {}_3p_x + \cdots\right) \\
\approx&\frac{m+1}{2m}\ddot{a}^{\star}_x + \frac{m-1}{2m}\frac{1}{1+r}\left(\ddot{a}^{\star}_x - 1\right),
\end{aligned}$$

in which $\ddot{a}^{\star}_x$ equals $\ddot{a}_x$ evaluated at the effective rate of interest $\frac{1+i}{1+r} - 1$.

7. Potential cashflows are shown below.



Premium (£)	$-P$				
Annuity (£)	1,000	1,000	1,000	1,000	$\cdots$
Expenses (£)	$0.05P + 150$	150	150	150	$\cdots$

The premium basis is as follows.

Interest:	6% per annum effective.
Mortality:	AM92 Select.
Expenses:	5% of the premium, and £150 at the time of each annuity payment.

(a) The EPV of the annuity payments is

$$12,000\left(\ddot{a}^{12}_{\overline{5}|} + {}_5|\ddot{a}^{(12)}_{[65]}\right).$$

As $(1 - \frac{d^{(12)}}{12})^{-12} = 1 + i$,

$$\ddot{a}^{12}_{\overline{5}|} = \frac{1 - v^5}{d^{(12)}} = \frac{1 - (1.06)^{-5}}{12\left(1 - 1.06^{-\frac{1}{12}}\right)} = \frac{1 - (1.06)^{-5}}{0.058128} = 4.348047$$

and, applying Woolhouse's formula to the monthly annuity,

$$
\begin{aligned}
{}_{5|}\ddot{a}^{(12)}_{[65]} = v^5\,{}_5p_{[65]}\,\ddot{a}^{(12)}_{70} = (1.06)^{-5}\,\frac{\ell_{70}}{\ell_{[65]}}\left(\ddot{a}_{70}-\frac{11}{24}\right) &= (1.06)^{-5}\,\frac{8054.0544}{8821.2612}\left(9.140-\frac{11}{24}\right)\\
&= (0.682267)\;(8.68167)\\
&= 5.92322,
\end{aligned}
$$

the EPV of the annuity payments is

$$12,000\;(4.348047+5.92322) = 12,000\;(10.271267) = £123,255.20.$$

(b) The EPV of the expenses is

$$0.05P + (12\times 150)\left(\ddot{a}^{12}_{\overline{5}|} + {}_{5|}\ddot{a}^{(12)}_{[65]}\right) = 0.05P + 1800\,(10.271267) = 0.05P + 18,488.28.$$

(c) Equating the premium (i.e. the money coming into the insurance company) with the EPV of the benefits and expenses (i.e. the money paid out by the insurance company), we get

$$P = 123,255.20 + 0.05P + 18,488.281 \quad\Rightarrow\quad P = \frac{123,255.20+18,488.28}{1-0.05} = £149,203.66.$$

8. Potential cashflows are shown below.

	45	46	47	$\cdots$	64	65	66	$\cdots$
Premium (£)	$-P$	$-P$	$-P$	$\cdots$	$-P$			
Annuity (£)						4,000	4,000	$\cdots$
Return of premium (£)		P	$2P$	$\cdots$	$20P$			
Expenses (£)	$0.02P$	$0.02P$	$0.02P$	$\cdots$	$0.02P$	10	10	$\cdots$

Time

(a) The EPV of the annuity payments is $4,000\,{}_{20|}\ddot{a}_{45}$. As

$${}_{20|}\ddot{a}_{45} = v^{20}\,{}_{20}p_{45}\,\ddot{a}_{65} = (1.04)^{-20}\,\frac{\ell_{65}}{\ell_{45}}\,\ddot{a}_{65} = (1.04)^{-20}\,\frac{8821.2612}{9801.3123}\,(12.276) = 5.042387,$$

we get $4,000\,{}_{20|}\ddot{a}_{45} = 4,000\,(5.042387) = £20,169.57$.

(b)

$$\begin{aligned}(IA)^{1}_{45:\overline{20|}} =&(IA)_{45} - v^{20}\,{}_{20}p_{45}\,((IA)_{65} + 20A_{65})\\ =&8.33628 - (1.04)^{-20}\,\frac{8821.2612}{9801.3123}\,(7.89442 + 20(0.52786))\\ =&8.33628 - (0.410752)\,(18.45162)\\ =&0.75724.\end{aligned}$$

(c) The EPV of the return-of-premium benefit in terms of P is

$$\begin{aligned}&Pvq_{45} + 2Pv^2\,{}_{1|}q_{45} + 3Pv^3\,{}_{2|}q_{45} + \cdots + 20Pv^{20}\,{}_{19|}q_{45}\\ =&P\left(vq_{45} + 2v^2\,{}_{1|}q_{45} + 3v^3\,{}_{2|}q_{45} + \cdots + 20v^{20}\,{}_{19|}q_{45}\right)\\ =&P\,(IA)^{1}_{45:\overline{20|}}\\ =&0.75724P.\end{aligned}$$

(d) The EPV of the expenses in terms of P is

$$0.02P\,\ddot{a}_{45:\overline{20|}} + 10\,{}_{20|}\ddot{a}_{45} = 0.02P(13.780) + 10(5.042387) = 0.2756P + 50.42387.$$

(e) Equating the EPV of the premiums (i.e. the money coming into the insurance company),

$$P\,\ddot{a}_{45:\overline{20|}} = 13.780P,$$

with the EPV of the benefits and expenses (i.e. the money paid out by the insurance company),

$$20,169.57 + 0.75724P + 0.2756P + 50.42387,$$

we get

$$13.780P = 20,169.57 + 0.75724P + 0.2756P + 50.42387.$$

Solving for P gives

$$P = \frac{20,169.57 + 50.42387}{13.780 - 0.75724 - 0.2756} = \frac{20,219.99387}{12.74716} = £1,586.24.$$